Your `Other_terms.tex`: the μ coefficients, DGLAP from Group II Row IV, the finite terms, and the running coupling

Your file lists, row by row, what is left after the log $\frac{k^+}{\Lambda}$ (JIMWLK) and log $\frac{\vee}{k^+}$ (BFKL) terms are taken out. Groups I and III are with c.c., Group II is without. Blue = new, red = removed. Only $\vee > k^+$ and $p^+ > \Lambda$ are used. $\vee$ is kept exact, and $\Lambda \to 0$ is taken only where the limit exists term by term.

Your file is the checked list. After removing the markup, your file is identical, equation by equation, to Secs. 2–4 (the remaining terms) of `Remaining_RC_DGLAP_finite.pdf` (`diff_check.py`). The differences are only:

- Group I Row V $d^3N_{2b\,\mathrm{UV}}$ keeps $2\log(1-\frac{\Lambda}{k^+})$, and $d^3N_{2b\,\mathrm{finite}}$ keeps Λ in two limits. Both go to 0 for $\Lambda \to 0$ (Sec. 4).
- $d^3N_{2b\,\mathrm{finite}}$ writes the $\frac{11}{6}$ line with $+$, where the earlier file had $-[-\frac{11}{6}]$; the two are equal.

So everything below uses your equations as they are.

1 Notation

- $a \equiv \frac{k^+}{\vee}$, $\beta_0 = \frac{11}{3}N_c$ ($n_f = 0$), $\alpha_s = \frac{g^2}{4\pi}$, $d_\perp = 2-\epsilon$.
- $c_0 \equiv 2e^{-\gamma_E}$, $\ell_r(\mu) \equiv \log\frac{r^2\mu^2}{c_0^2} = 2\gamma_E + \log\frac{r^2\mu^2}{4}$, $\hat{E}_r \equiv \frac{2}{\epsilon} + \ell_r(\mu)$.
- $X = x - y$, $X' = x' - y'$, $\Delta = y' - y$; $\mu \equiv \mu_{\overline{MS}}$.

LO in your notation, and LO with a weight f in the integrand:

$$\mathrm{LO}[f] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\, f\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\,\Big[U^{e'c}(y') - U^{e'c}(x')\Big]\,[U^{ec}(y) - U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x), \tag{1.1}$$

with $\mathrm{LO} \equiv \mathrm{LO}[1]$. $\mathrm{LO}_\xi[f]$ is the same with $e^{-ik\cdot(y'-y)} \to e^{-ik\cdot(y'-y)\frac{1}{\xi}}$. Since LO at momentum $(p^+, \mathbf{p})$ carries $\frac{1}{p^+}e^{-i\mathbf{p}\cdot(y'-y)}$,

$$\mathrm{LO}_\xi = \frac{1}{\xi}\,\frac{d^3N_{\mathrm{LO}}}{d^3k}\Big(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi}\Big), \qquad d^3k \equiv dk^+ d^2\mathbf{k}. \tag{1.2}$$

Your H_4 (the line that you write with $(1-\frac{k^+}{\vee})$):

$$\begin{aligned} H_4 \equiv \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2} \\ \times\Big[U^{e'c}(z) - U^{e'c}(x')\Big]\,[U^{ec}(z) - U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x). \end{aligned} \tag{1.3}$$

Splitting functions.

$$P^{(r)}_{gg,+}(\xi) = 2N_c\Big[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\Big], \qquad P_{gg}(\xi) = P^{(r)}_{gg,+}(\xi) + \frac{\beta_0}{2}\,\delta(1-\xi). \tag{1.4}$$

Here $\frac{1}{(1-\xi)_+}$ is the plus distribution of $[0,1]$. Cut at $\xi \geq a$ it acts as

$$\int_a^1 d\xi\,\frac{g(\xi)}{(1-\xi)_+} = \int_a^1 d\xi\,\frac{g(\xi)-g(1)}{1-\xi} + g(1)\log(1-a). \tag{1.5}$$

Your prescription has no $\log(1-a)$: $\int\frac{d\xi}{\xi^2}\frac{\xi^2}{(1-\xi)_+}G = \int\frac{d\xi}{1-\xi}[G(\xi)-G(1)]$. On $[a,1]$ the two are related exactly by (`checks.py`, item 3)

$$\int_a^1\frac{d\xi}{\xi^2}\Big[\xi + \frac{\xi^2}{(1-\xi)_+}\Big]_{\mathrm{yours}} G(\xi) = \int_a^1\frac{d\xi}{\xi^2}\,\frac{\xi}{(1-\xi)_+}\,G(\xi) \;-\; \log(1-a)\,G(1). \tag{1.6}$$

2 (1) All coefficients of μ

Where μ appears. In your file μ appears in five places: the two orange brackets (Group I Row II part 1; Group I Row V $d^3N_{2b\,\mathrm{UV}}$) and lines (L1), (L2), (L4) of Group II Row IV. Nowhere else: the $d_\perp$ of Group I Row V 2a, 2b finite, Group II Row I d^3N_{2f} and Group II Row IV (L5) multiplies finite integrands, so $d_\perp \to 2$ there exactly for $\epsilon \to 0$. Every μ comes in the form $\hat{E}_r = \frac{2}{\epsilon} + \ell_r(\mu)$, i.e. from $(\mu^2)^{\epsilon/2}/\epsilon$. So **the coefficient of $\log\mu^2$ equals the coefficient of $\frac{2}{\epsilon}$**, source by source.

Step 1: Group I Row II, part 1. The colour factor with $w \to y$, $w' \to y'$ and $a \to c$, $b' \to e'$, $b \to e$ ($U^{bc} = U^{cb}$):

$$\Big[U^{ab'}(x')\rho^{b'}(x') - U^{ab'}(y')\rho^{b'}(x')\Big]\Big[U^{ab}(y)\rho^{b}(x) - U^{ab}(x)\rho^{b}(x)\Big] = -\Big[U^{e'c}(y') - U^{e'c}(x')\Big]\big[U^{ec}(y) - U^{ce}(x)\big]\,\rho^{e'}(x')\rho^{e}(x).$$

The prefactor is $\frac{g^4 N_c}{32\pi^4} = \frac{g^2}{2\pi^2}\cdot\frac{\alpha_s N_c}{4\pi}$; the measure, the phase and the kernel become those of (1.1); your orange bracket is $-\hat{E}_{x-w} \to -\hat{E}_X$. The c.c. is the same with $X \to X'$. So

$$\left.\frac{d^3N_1}{d^3k}\right|_{\text{G1 II}} + \text{c.c.} = \frac{\alpha_s N_c}{4\pi}\Big[\frac{11}{3} + \log\Big(1-\frac{k^+}{\vee}\Big)\Big]\text{LO}\big[\hat{E}_X + \hat{E}_{X'}\big] + \frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9} - \frac{\pi^2}{3}\Big]\text{LO}$$
$$- \frac{\alpha_s}{2\pi}N_c \log\Big(1-\frac{k^+}{\vee}\Big)\log\Big(\frac{\vee}{k^+} - 1\Big)\text{LO}. \tag{2.1}$$

The last term is $\log^2(1-\frac{k^+}{\vee}) + \log(1-\frac{k^+}{\vee})\log\frac{\vee}{k^+} = \log(1-\frac{k^+}{\vee})\log(\frac{\vee}{k^+}-1)$.

Step 2: Group I Row V, $d^3N_{2b\,\text{UV}}$. With $y \to x$, $z \to y$, $w' \to y'$ the colour factor is $+$ that of LO. The prefactor is $\frac{g^4 N_c}{16\pi^4} = \frac{g^2}{2\pi^2}\cdot\frac{\alpha_s N_c}{2\pi}$, and the orange bracket is $+\hat{E}_{y-z} \to \hat{E}_X$:

$$\frac{d^3N_{2b\,\text{UV}}}{d^3k} + \text{c.c.} = \frac{\alpha_s N_c}{2\pi}\Big[2\log\Big(1-\frac{\Lambda}{k^+}\Big) - \frac{11}{6}\Big]\text{LO}\big[\hat{E}_X + \hat{E}_{X'}\big] \xrightarrow{\Lambda\to 0} -\frac{\alpha_s\beta_0}{4\pi}\text{LO}\big[\hat{E}_X + \hat{E}_{X'}\big]. \tag{2.2}$$

Step 3: Group II Row IV, (L1) and (L4). $\frac{g^2}{4\pi^2} = \frac{\alpha_s}{\pi}$, and your bracket is

$$-\frac{d_\perp}{\epsilon} - d_\perp\gamma_E - \frac{d_\perp}{2}\log\frac{\Delta^2\mu^2}{4} - 1 = -\frac{2}{\epsilon} + 1 - 2\gamma_E - \log\frac{\Delta^2\mu^2}{4} - 1 + O(\epsilon) = -\hat{E}_\Delta + O(\epsilon). \tag{2.3}$$

The -1 cancels the $+1$ of $-\frac{d_\perp}{\epsilon}$. With (1.6), your (L4) is exactly the $\log(1-a)$ term of the standard plus distribution (1.5):

$$\text{(L1)} + \text{(L4)} = \frac{\alpha_s N_c}{\pi}\int_a^1 \frac{d\xi}{\xi^2}\Big[\xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{(1-\xi)_+}\Big]\text{LO}_\xi\big[-\hat{E}_\Delta\big] = \frac{\alpha_s}{2\pi}\int_a^1 \frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,\text{LO}_\xi\big[-\hat{E}_\Delta\big]. \tag{2.4}$$

Step 4: Group II Row IV, (L2). (L2) has $U(y)$ in place of $U(y')$ and the kernel $\frac{(x-y)\cdot(x'-y)}{(x-y)^2(x'-y)^2}$. So y' appears only in the phase and in the log:

$$\int_{y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\log\frac{(y'-y)^2\mu^2}{4} = -\frac{4\pi\xi^2}{\mathbf{k}^2} + \log\frac{\mu^2}{4}\,(2\pi)^2\xi^2\,\delta^{(2)}(\mathbf{k}) \qquad (\texttt{checks.py}\text{, item 4}). \tag{2.5}$$

For $\mathbf{k} \neq 0$ the μ term is zero: **(L2) has no μ.** (L2) has no pole either, so $d_\perp \to 2$ in it.

Step 5: the counterterms. In $\overline{MS}$, the coupling of LO renormalized at μ_R and the fragmentation function at μ_F add (Sec. 3, Step 6)

$$-\frac{\alpha_s\beta_0}{4\pi}\Big[\frac{2}{\epsilon} + \log\frac{\mu^2}{\mu_R^2}\Big]\text{LO} \quad \text{(coupling)}, \qquad +\frac{\alpha_s}{2\pi}\Big[\frac{2}{\epsilon} + \log\frac{\mu^2}{\mu_F^2}\Big]\int_a^1\frac{d\xi}{\xi^2}P_{gg}(\xi)\,\text{LO}_\xi \quad \text{(fragmentation)}. \tag{2.6}$$

The table. Coefficient of $\log\mu^2$, which is also the coefficient of $\frac{2}{\epsilon}$, in units of $\frac{\alpha_s}{2\pi}$. Here $P\otimes\text{LO} \equiv \int_a^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,\text{LO}_\xi$ (`checks.py`, item 1):

source	coefficient of $\log\mu^2$ (and of $\frac{2}{\epsilon}$)	what it is
G1 Row II part 1 + c.c., (2.1)	$N_c\big[\frac{11}{3} + \log(1-a)\big]\text{LO} = \big[\beta_0 + N_c\log(1-a)\big]\text{LO}$	β_0; Q1
G1 Row V $2b$ UV + c.c., (2.2)	$2N_c\big[2\log(1-\frac{\Lambda}{k^+}) - \frac{11}{6}\big]\text{LO} \to -\beta_0\,\text{LO}$	β_0
G2 Row IV (L1)+(L4), (2.4)	$-P\otimes\text{LO}$	DGLAP
G2 Row IV (L2)	0	—
G2 Row IV (L3), (L5); all other rows	0	—
sum of your file	$N_c\log(1-a)\,\text{LO} - P\otimes\text{LO}$ $\quad(+4N_c\log(1-\frac{\Lambda}{k^+})\text{LO}\to 0)$	
coupling counterterm	$-\frac{\beta_0}{2}\text{LO}$	
fragmentation counterterm	$+P\otimes\text{LO} + \frac{\beta_0}{2}\text{LO}$	
total	$N_c\log(1-a)\,\text{LO}$	Q1

- The two β_0 entries cancel exactly: pole, γ_E, μ, and also $\log(x-w)^2$ against $\log(y-z)^2$, since (2.1) and (2.2) have the same $\text{LO}[\hat{E}_X + \hat{E}_{X'}]$.
- The DGLAP entry $-P\otimes\text{LO}$, with the $\frac{\beta_0}{2}$ of the coupling counterterm, is $-P_{gg}\otimes\text{LO}$, and it is removed by the fragmentation counterterm.
- **What is left is $\frac{\alpha_s}{2\pi}N_c\log(1-\frac{k^+}{\vee})\log\mu^2\,\text{LO}$, together with its pole** (Q1, Eq. (6.1)). It comes from the cut $\xi \geq \frac{k^+}{\vee}$ in the virtual term of Group I Row II. It vanishes for $\vee \to \infty$ at fixed k^+, but it is not zero at finite $\vee$ (Sec. 6).
- After the counterterms, the scales that remain are μ_R and μ_F: $\frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}\text{LO}$ (Sec. 5) and $-\frac{\alpha_s}{2\pi}P\otimes\text{LO}[\ell_\Delta(\mu_F)]$ (Sec. 3).

3 (2) DGLAP from Group II Row IV, step by step

Step 0: your row. Your five lines are (L1); (L2), the subtraction at $y'=y$; (L3), the H_4 line; (L4), the $\log(1-\frac{k^+}{\vee})$ line; and (L5), the linear-polarization line:

$$
\begin{aligned}
(L1)\quad & \frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\\
&\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\xi+\frac{\xi^2}{(1-\xi)_+}\right]\right]\times\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
(L2)\quad & -\frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y)}{(x-y)^2\,(x'-y)^2}\left[-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)\right]\\
&\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(1-\xi)}{\xi}\right]\right]\times\Big[U^{e'c}(y)-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
(L3)\quad & -\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\
&\Big[U^{e'c}(z)-U^{e'c}(x')\Big]\times\left[U^{ec}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
(L4)\quad & +\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.\\
&\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu_{\overline{MS}}^2}{4}\right)-1\right]\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
(L5)\quad & +\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\times\\
&\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)
\end{aligned}
\tag{3.1}
$$

Step 1: (L1) in units of LO. $\frac{g^2}{4\pi^2}=\frac{\alpha_s}{\pi}$, and the x,y,x',y' integrand of (L1) is that of (1.1) with $e^{-ik\cdot(y'-y)}\to e^{-ik\cdot(y'-y)\frac{1}{\xi}}$. So

$$
\text{(L1)}=\frac{\alpha_s N_c}{\pi}\int_a^1\frac{d\xi}{\xi^2}\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\xi+\frac{\xi^2}{(1-\xi)_+}\Big]_{\text{yours}}\text{LO}_\xi\big[\mathcal{B}_\Delta\big],\qquad \text{(L4)}=\frac{\alpha_s N_c}{\pi}\log(1-a)\,\text{LO}\big[\mathcal{B}_\Delta\big],
\tag{3.2}
$$

where $\mathcal{B}_\Delta$ is your bracket (2.3).

Step 2: (L4) completes the plus distribution. By (1.6), the bracket of (L1) is the standard one minus $\log(1-a)\,\delta(1-\xi)$, and $\text{LO}_{\xi=1}=\text{LO}$. (L4) adds this term back:

$$
\text{(L1)}+\text{(L4)}=\frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,\text{LO}_\xi\big[\mathcal{B}_\Delta\big].
\tag{3.3}
$$

This is needed in Step 5. Without (L4) the pole does not factorize (`checks.py`, item 5: 30% and 110% mismatch).

Step 3: the pole. By (2.3), $\mathcal{B}_\Delta=-\frac{2}{\epsilon}-\ell_\Delta(\mu)+O(\epsilon)$. The $O(\epsilon)$ multiplies a finite integral and drops out. Write $\ell_\Delta(\mu)=\ell_\Delta(\mu_F)+\log\frac{\mu^2}{\mu_F^2}$:

$$
\text{(L1)}+\text{(L4)}=\frac{\alpha_s}{2\pi}\Big[-\frac{2}{\epsilon}-\log\frac{\mu^2}{\mu_F^2}\Big]\int_a^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,\text{LO}_\xi\;-\;\frac{\alpha_s}{2\pi}\int_a^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,\text{LO}_\xi\big[\ell_\Delta(\mu_F)\big].
\tag{3.4}
$$

Step 4: the convolution. By (1.2),

$$
\int_a^1\frac{d\xi}{\xi^2}\,P(\xi)\,\text{LO}_\xi=\int_{k^+/\vee}^1\frac{d\xi}{\xi^3}\,P(\xi)\,\frac{d^3N_{\text{LO}}}{d^3k}\Big(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi}\Big).
\tag{3.5}
$$

This describes a gluon of momentum $(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi})$, produced at LO, that turns into a gluon with the fraction ξ of its momentum. The lower limit says that the parent has $\frac{k^+}{\xi}\le\vee$. This is the structure of a fragmentation function.

Step 5: fragmentation. For a hadron h ($d^3k=dk^+d^2\mathbf{k}$, all components scaled by ζ):

$$
\frac{d^3N_h}{d^3k}=\int_{k^+/\vee}^1\frac{d\zeta}{\zeta^3}\,D^{\text{bare}}_{h/g}(\zeta)\,\frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big).
\tag{3.6}
$$

For the gluon of momentum $\frac{k}{\zeta}$ the cut in Step 4 is $\xi\ge\frac{a}{\zeta}$. Insert (3.4) and change variables $z=\zeta\xi$. The region $a\le\zeta\le1$, $\frac{a}{\zeta}\le\xi\le1$ is $a\le z\le\zeta\le1$, and $\frac{d\zeta}{\zeta^3}\frac{d\xi}{\xi^3}=\frac{dz}{z^3}\frac{d\zeta}{\zeta}$. Then

$$
\int_a^1\frac{d\zeta}{\zeta^3}D(\zeta)\int_{a/\zeta}^1\frac{d\xi}{\xi^3}P(\xi)\frac{d^3N_{\text{LO}}}{d^3k}\Big(\frac{k}{\zeta\xi}\Big)=\int_a^1\frac{dz}{z^3}\,\frac{d^3N_{\text{LO}}}{d^3k}\Big(\frac{k}{z}\Big)\,[P\otimes D](z),\qquad [P\otimes D](z)=\int_z^1\frac{d\zeta}{\zeta}\,P\Big(\frac{z}{\zeta}\Big)D(\zeta).
\tag{3.7}
$$

The cut $\xi \geq \frac{a}{\zeta}$ becomes $z \geq a$, independent of ζ. So $[P \otimes D]$ is the ordinary DGLAP convolution, with the plus distribution of $[0,1]$. This holds only if the cut plus distribution is (1.5), which is why (L4) belongs here (`checks.py`, item 5: 2×10^{-12}).

Step 6: $\overline{MS}$ fragmentation function. Add the coupling counterterm (2.6). Its pole is $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon})\frac{\beta_0}{2}\mathrm{LO}$, the $\delta(1-\xi)$ part of P_{gg}; the β_0 of your Group I rows cancels by itself (Sec. 5). The pole of (3.4) becomes $\frac{\alpha_s}{2\pi}(-\frac{2}{\epsilon} - \log\frac{\mu^2}{\mu_F^2})P_{gg}\otimes \mathrm{LO}$. In (3.6) it is removed by

$$D_{h/g}^{\mathrm{bare}}(z) = D_{h/g}(z,\mu_F) + \frac{\alpha_s}{2\pi}\Big[\frac{2}{\epsilon} + \log\frac{\mu^2}{\mu_F^2}\Big]\big[P_{gg}\otimes D_{h/g}\big](z,\mu_F). \tag{3.8}$$

- D^{bare} does not depend on μ_F, so

$$\mu_F^2\frac{\partial D_{h/g}(z,\mu_F)}{\partial \mu_F^2} = \frac{\alpha_s}{2\pi}\big[P_{gg}\otimes D_{h/g}\big](z,\mu_F). \tag{3.9}$$

  This is the DGLAP equation.

- Its $\frac{\beta_0}{2}$ part, with the coupling counterterm, leaves the renormalization-group term $\frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}\mathrm{LO}$ (Sec. 5).

Step 7: the DGLAP part of the row. The DGLAP part of Group II Row IV is

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\mathrm{DGLAP\ pole}} = \frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon} - \log\frac{\mu^2}{\mu_F^2}\Big)\int_{k^+/\vee}^1 \frac{d\xi}{\xi^3}\,P_{gg,+}^{(r)}(\xi)\,\frac{d^3N_{\mathrm{LO}}}{d^3k}\Big(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi}\Big).} \tag{3.10}$$

With the $\frac{\beta_0}{2}$ of the coupling counterterm it is absorbed into $D_{h/g}(\zeta,\mu_F)$ through (3.8). Its finite partner $-\frac{\alpha_s}{2\pi}P_{gg,+}^{(r)}\otimes\mathrm{LO}[\ell_\Delta(\mu_F)]$ stays in the gluon cross section. Its μ_F dependence is cancelled by the DGLAP evolution of $D_{h/g}$.

Step 8: (L2)+(L3). The subtraction (L2) after the y' integral (Step 4 of Sec. 2; $\frac{d_\perp}{2}\to 1$):

- $\int_{y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\big[-\log\frac{(y'-y)^2\mu^2}{4}\big] = \frac{4\pi\xi^2}{\mathbf{k}^2}$;
- $\int_a^1 \frac{d\xi}{\xi^2}\frac{1-\xi}{\xi}\,\xi^2 = \log\frac{\vee}{k^+} - \big(1-\frac{k^+}{\vee}\big)$;
- $\int_{w,w'} e^{-ik\cdot(w'-w)}\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2} = \frac{(2\pi)^2}{\mathbf{k}^2}$.

With the prefactor ratio 1 (`checks.py`, item 2) and $y\to z$,

$$(\mathrm{L2}) = -\Big[\log\frac{\vee}{k^+} - \Big(1-\frac{k^+}{\vee}\Big)\Big]H_4, \qquad (\mathrm{L3}) = -\Big(1-\frac{k^+}{\vee}\Big)H_4, \qquad \boxed{(\mathrm{L2})+(\mathrm{L3}) = -\log\frac{\vee}{k^+}\,H_4.} \tag{3.11}$$

The BFKL list has $+\log\frac{\vee}{k^+}H_4$ for this row (`BFKL_check.pdf`, C2). (L2)+(L3) removes from (L1) the $\log\frac{\vee}{k^+}$ of its small-ξ end, where $e^{-ik\cdot(y'-y)/\xi}$ forces $y'\to y$. So the finite part of (L1)+(L4) together with (L2)+(L3) stays finite for $\vee\to\infty$.

The remaining terms of Group II Row IV. After the DGLAP pole is absorbed in $D_{h/g}(\zeta,\mu_F)$ (Step 7), and with $\epsilon\to 0$ exact in what is finite, your row is

$$\begin{aligned}
\frac{d^3N_{\mathrm{rem}}^{\mathrm{G2\,IV}}}{d^3k} &= \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\Big[-2\gamma_E - \log\Big(\frac{(y'-y)^2\,\mu_F^2}{4}\Big)\Big]\\
&\quad\times\Big[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\frac{1}{\xi}}\Big[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\xi+\frac{\xi^2}{(1-\xi)_+}\Big] + \log\Big(1-\frac{k^+}{\vee}\Big)e^{-ik\cdot(y'-y)}\Big]\\
&\quad\times\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\,[U^{ec}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\\
&\quad-\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y)}{(x-y)^2\,(x'-y)^2}\Big[-\log\Big(\frac{(y'-y)^2\,\mu^2}{4}\Big)\Big]\\
&\quad\times\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(1-\xi)}{\xi}\Big[U^{e'c}(y)-U^{e'c}(x')\Big]\,[U^{ec}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\\
&\quad-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\Big(1-\frac{k^+}{\vee}\Big)\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\
&\quad\times\Big[U^{e'c}(z)-U^{e'c}(x')\Big]\,[U^{ec}(z)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x)\\
&\quad+\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\frac{1}{\xi}}\\
&\quad\times\Big[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\Big]\\
&\quad\times\Big[U^{e'c}(y')-U^{e'c}(x')\Big]\,[U^{ec}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^{e}(x).
\end{aligned} \tag{3.12}$$

- Lines 1–2 are (L1)+(L4) without the pole, at μ_F: $-\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2} P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi[\ell_\Delta(\mu_F)]$. Your bracket $[\xi + \frac{\xi^2}{(1-\xi)_+}]$ and the $\log(1-\frac{k^+}{\vee})$ term are kept as you wrote them.
- Lines 3–6 are (L2) and (L3), $= -\log\frac{\vee}{k^+} H_4$. The μ in (L2) drops out (Step 4 of Sec. 2).
- Lines 7–8 are (L5) with $d_\perp = 2$. They are finite and not DGLAP: the Δ angular average of the bracket is zero, and the Fourier transform gives the linearly polarized structure $\frac{2k^ik^j}{\mathbf{k}^2} - \delta^{ij}$. They are also finite for $\vee \to \infty$: the measure is $d\xi$, not $\frac{d\xi}{\xi^2}$, so the small-ξ end behaves as $\int d\xi\,\xi(1-\xi)$.

4 (3) The other finite terms of the cross section

Approximations: none. $\vee$ is exact. The only limits are $\Lambda \to 0$ in the two limits of $d^3N_{2b\,\mathrm{finite}}$ and in $2\log(1-\frac{\Lambda}{k^+})$, and $\epsilon \to 0$ in integrands without poles ($d_\perp \to 2$). Both limits exist term by term: the integrands of $d^3N_{2b\,\mathrm{finite}}$ are regular at $p^+, q^+ \to 0, k^+$, the first through the plus prescription and the second because $[(k^+-q^+)(y-x)^2 + q^+(y-z)^2]^{-1}$ is finite there. Your equations are copied unchanged.

4.1 From Group I Row II part 1 (with c.c.)

After the β_0 term (cancelled, Sec. 5) and Q1 (Eq. (6.1)), (2.1) leaves two finite terms:

$$\begin{aligned}
\frac{d^3N_{\mathrm{const}}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\Big[-\frac{67}{9}+\frac{\pi^2}{3}\Big] \\
&\quad\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[U^{ab}(w)\,\rho^b(x) - U^{ab}(x)\,\rho^b(x)\Big] + \text{c.c.} = \frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\mathrm{LO}, \qquad (4.1)\\
\frac{d^3N_{V_\vee}}{d^3k} &= \frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\,\log\Big(1-\frac{k^+}{\vee}\Big)\log\Big(\frac{\vee}{k^+}-1\Big) \\
&\quad\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[U^{ab}(w)\,\rho^b(x) - U^{ab}(x)\,\rho^b(x)\Big] + \text{c.c.} \\
&= -\frac{\alpha_s}{2\pi}N_c\log\Big(1-\frac{k^+}{\vee}\Big)\log\Big(\frac{\vee}{k^+}-1\Big)\mathrm{LO}. \qquad (4.2)
\end{aligned}$$

$V_\vee$ is bounded for every $\vee > k^+$ and vanishes for $\vee \to \infty$.

4.2 Group I Row II part 2, d^3N_{2b} (with c.c.)

Unchanged (three ρ's, no large logarithm):

$$\begin{aligned}
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\frac{g^4}{64\pi^6}\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y} e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,R^i(\mathbf{k},\mathbf{p}) \\
&\quad\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{hbc}\,U^{ah}(w)\Big\{\rho^b(x),\rho^c(y)\Big\} - f^{abc}\,U^{bd}(x)\,U^{ce}(y)\Big\{\rho^d(x),\rho^e(y)\Big\}\Big] \qquad (4.3)
\end{aligned}$$

where

$$R^i = -2k^i\log\left(\frac{p^2}{k^2+\frac{k^+}{\vee}p^2}\right) + \frac{(2p^2-2\,k\cdot p+k^2)\,k^i + k^2p^i}{(k-p)^2}\log\left[\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right)\right] - \frac{p^2k^i+k^2p^i}{(k-p)^2}\log\left(1-\frac{k^+}{\vee}\right). \qquad (4.4)$$

4.3 Group I Row III (with c.c.)

$$\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&-\frac{ig^4\,f^{abd}}{16\pi^5}\frac{1}{k^+}\log\Big(1-\frac{k^+}{\vee}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{[(x-z)^2+(x-w)^2]}{[(x-z)^2-(x-w)^2]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{[(x-w)^2+(x-z)^2]}{[(x-z)^2-(x-w)^2]^2}\times\log\left[\frac{(x-z)^2-\frac{k^+}{\vee}(x-w)^2}{(x-z)^2-(x-w)^2}\right] \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{[(x-w)^2+(x-z)^2]}{[(x-z)^2-(x-w)^2]^2}\times\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right] \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\log\left[\frac{\frac{k^+}{\vee}(x-w)^2-(x-z)^2}{(x-w)^2}\right] \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\} \qquad (4.5)
\end{aligned}$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\}
\end{aligned}
\tag{4.6}
$$

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = &\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}_0\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}_0\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}_0\int_{x,y,z,x'}\frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\}
\end{aligned}
\tag{4.7}
$$

$$
\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\Bigg\{
&-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}_0\int_{x,y,z,x'}\frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&-\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\,\log\left[\frac{(z-w)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\,\log\left[\frac{(z-w)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\}
\end{aligned}
\tag{4.8}
$$

where

$$
\begin{aligned}
\mathcal{A}_0=\frac{1}{\sqrt{(x-w)^2\,(x-z)^2-\left[(x-w)\cdot(x-z)\right]^2}}\Bigg\{&\arctan\left[\frac{(x-z)^2-(x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2-\left[(x-w)\cdot(x-z)\right]^2}}\right]\\
&+\arctan\left[\frac{(x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2-\left[(x-w)\cdot(x-z)\right]^2}}\right]\Bigg\}
\end{aligned}
\tag{4.9}
$$

4.4 Group I Row IV (with c.c.)

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik\cdot(w'-w)}\Bigg\{
&\frac{ig^4}{8\pi^5}\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.\\
&\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}+\frac{\delta^{ij}}{2\,(z-w)^2}\right]\\
&+\frac{ig^4}{8\pi^5}\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big]
\end{aligned}
\tag{4.10}
$$

4.5 Group I Row V (with c.c.)

First part.

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k}=&\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\\
&\left[\delta_{im}\,\delta_{jk}-\left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\\
&\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^{e}(y),\rho^{b}(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^{d}(y),\rho^{e}(y')\}\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,z,y,x',y'w'}e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\left[\frac{1}{q^+}\right]_+\\
&\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^{e}(y),\rho^{b}(y')\}-f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^{d}(y),\rho^{e}(y')\}\Big]
\end{aligned}
\tag{4.11}
$$

Second part, d^3N_{2a}.

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\Bigg]\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\int_{k^+}^0\frac{dq^+}{2\pi}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left\{\left[\frac{k^+}{q^+}\right]_+\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{4.12}
$$

Second part, $d^3N_{2b\,\mathrm{UV}}$. Nothing is left: $-\frac{11}{6}$ is the β_0 term, which cancels that of Row II part 1 (Sec. 5), and $2\log(1-\frac{\Lambda}{k^+})\to 0$.

Second part, $d^3N_{2b\,\mathrm{finite}}$ ($\Lambda\to 0$ in the limits):

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+(k^+-p^+)}{k^{+2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\,\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\right]\\
&\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\,\left[\frac{k^+}{q^+}\right]_+\times\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{11}{6}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{4.13}
$$

The last line (with $\frac{11}{6}$) is the subtraction of the UV counterterm, written with the kernel $\frac{(y-x)\cdot(y-z)}{(y-x)^2(x-z)^2}$. Alone it diverges logarithmically at $x\to z$. Only the sum of all lines of $d^3N_{2b\,\mathrm{finite}}$ is finite, because the $\frac{1}{(x-z)^2}$ terms cancel for every p^+ (`rowV/Part2b_MSbar.pdf`). Keep the lines together.

4.6 Group II Row I

$$\frac{d^3N_{1a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times$$
$$\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}$$
$$\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \quad (4.14)$$

$$\frac{d^3N_{1b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{$$
$$\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w')^2}{(x-w')^2}\right]\times$$
$$\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]$$
$$+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}$$
$$\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right] \quad (4.15)$$

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\Bigg\{$$
$$\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times$$
$$\Bigg[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}$$
$$+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}$$
$$+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}$$
$$-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\Bigg]\times$$
$$\left[\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]-\log\left[\frac{(x-w)^2}{(x'-w')^2}\right]\right]\Bigg\}\times$$
$$\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\Bigg] \quad (4.16)$$

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2\right]}\times$$
$$\Bigg[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}$$
$$+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\Bigg]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times$$
$$\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\Bigg]$$

$$
- \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times
$$

$$
\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
$$

$$
\left.+\,\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
+\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\,-(x-w)^2\,(x'-z)^2\,(x'-w')^2]}\times
$$

$$
\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.
$$

$$
\left.+\,\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{4.17}
$$

$$
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.
$$

$$
\left.+\,\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
$$

$$
-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.
$$

$$
\left.+\,\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\left.-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{4.18}
$$

$$
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k}=+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}
$$

$$
\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.
$$

$$
\left.+\,\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times
$$

$$
\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times
$$

$$
\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
$$

$$
\begin{aligned}
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&+ \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-z)^2}{\left[(x'-w')^2-(x'-z)^2\right]}\, \frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]} \\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2} \\
&+ \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg]\times \\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big]
\end{aligned}
\tag{4.19}
$$

$$
\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-w)^2}{\left[(x-z)^2-(x-w)^2\right]}\, \frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]} \\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2} \\
&+ \frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times \\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2}{\left[(x'-w')^2-(x'-z)^2\right]}\, \frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]} \\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2} \\
&+ \frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times \\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big]
\end{aligned}
\tag{4.20}
$$

$$
\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]} \\
&\frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&+ \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \left[\frac{1}{\vee}-\frac{1}{k^+}\right] \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{1}{(z-w)^2\,(z-w')^2} \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x)
\end{aligned}
$$

$$
\left. - f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \right]
$$

$$
+ \frac{1}{(2\pi)^3}\, \frac{g^4\, d_\perp}{32\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}
$$

$$
\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\,\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right] \times
$$

$$
\left[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \right.
$$

$$
\left. - f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x) \right] \tag{4.21}
$$

4.7 Group II Row IV

Equation (3.12).

4.8 Group III (with c.c.)

Row I.

$$
\frac{d^3 N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3} \int_{x,y,x',z,w,w'} e^{-ik(w'-w)} \left\{
$$

$$
- \frac{ig^4}{8\pi^5}\, \frac{1}{k^+}\, \frac{(x'-w')^i (y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right] \times \left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.
$$

$$
\left. - \frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]
$$

$$
- \frac{ig^4}{8\pi^5}\, \frac{1}{k^+}\, \log\left(1-\frac{k^+}{\vee}\right) \frac{(x'-w')^i (y-z)^j}{(x'-w')^2\,(y-z)^2}\, \left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right] \Bigg\} \times
$$

$$
\left[U^{ab'}(x')\, U^{c'd'}(y)\, U^{c'b}(z)\, \rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)\, U^{d'e'}(y) U^{ac'}(w')\, \rho^{c'}(x')\rho^{e'}(y) \right] \times
$$

$$
\left[f^{cbd}\, U^{ac}(w)\, \rho^{d}(x) - f^{acd}\, U^{bc}(z)\, U^{de}(x)\, \rho^{e}(x) \right] \tag{4.22}
$$

Row II.

$$
\frac{d^3 N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3} \int_{x,y,x',z,w,w'} e^{-ik(w'-w)} \left\{
$$

$$
- \frac{ig^4}{8\pi^5}\, \frac{1}{k^+}\, \frac{(x'-w')^i (y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right] \times \left[\frac{\delta^{ij}}{2\,(z-w)^2} + \frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.
$$

$$
\left. - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right]
$$

$$
- \frac{ig^4}{8\pi^5}\, \frac{1}{k^+}\, \log\left(1-\frac{k^+}{\vee}\right) \frac{(x'-w')^i (y-z)^j}{(x'-w')^2\,(y-z)^2}\, \left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right] \Bigg\} \times
$$

$$
\left[U^{c'd'}(y)\, U^{ab'}(x')\, U^{bc'}(z)\, \rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\, \rho^{b}(y)\rho^{b'}(x') \right] \times \left[f^{cbd}\, U^{ac}(w)\, \rho^{d}(x) - f^{acd}\, U^{bc}(z)\, U^{de}(x)\, \rho^{e}(x) \right] \tag{4.23}
$$

Row III, first part.

$$
\frac{d^3 N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\, \frac{ig^4\, f^{c'd'a}}{8\pi^4}\, \frac{1}{k^+} \int_0^{\vee-k^+} \frac{dp^+}{2\pi} \int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\, e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\, \frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$
\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)} - \left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_+ + \frac{1}{(p^++k^+)}\right)\delta_{jm}\,\delta_{ik'}\right] \left[U^{e'c'}(y')\, U^{d'b}(z)\, \rho^{e'}(x') - U^{bd'}(z)\, U^{c'e'}(x')\, \rho^{e'}(x')\right] \times
$$

$$
\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\, U^{ad}(y)\, U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]
$$

$$
+ \frac{1}{(2\pi)^3}\, \frac{ig^4\, f^{c'd'a}}{8\pi^4}\, \frac{1}{k^+} \int_0^{\vee-k^+} \frac{dp^+}{2\pi} \int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\, e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\, \frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}
$$

$$
\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right] \left[\frac{(x'-y')^{k'}}{(x'-y')^2}\, \left[U^{e'c'}(y')\, U^{d'b}(z)\, \rho^{e'}(x') - U^{bd'}(z)\, U^{c'e'}(x')\, \rho^{e'}(x')\right]\right.
$$

$$
- \frac{(x'-z)^{k'}}{(x'-z)^2}\,\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\Bigg]\times
$$

$$
\Big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\Big]
$$

$$
- \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$
\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\Big] \tag{4.24}
$$

Row III, second part.

$$
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}
$$

$$
\Bigg[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)}\,\left[(x-z)^2-(x-w)^2\right] + \frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2} - \frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.
$$

$$
\left.- \frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right] + \frac{\delta_{k'm}}{(p^++k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2} + \frac{(x-z)\cdot(x-w)}{2\,(x-z)^2} - \frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]
$$

$$
- \left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_+ + \frac{1}{(p^++k^+)}\right)\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2} + \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]
$$

$$
- \frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2} - \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2} + \frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2} + \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\Bigg]\times
$$

$$
\Big[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]
$$

$$
+ \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m}{(y'-z)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}
$$

$$
\left[-\frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times
$$

$$
\Bigg[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\Big[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]
$$

$$
- \frac{(x'-z)^{k'}}{(x'-z)^2}\,\Big[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\Bigg]\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]
$$

$$
+ \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-z)^{k'}}{(w'-z)^2\,(x'-z)^2}
$$

$$
\left[\frac{(x-w)^2}{(x-z)^2\,(p^++k^+)\,[p^+\,(x-z)^2+k^+\,(x-w)^2]}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times
$$

$$
\Big[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]
$$

$$
- \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.
$$

$$
\left.+ \frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\Big[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times
$$

$$
\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big] \tag{4.25}
$$

Row V.

$$
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}\,(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$
\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)} - \left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_+ + \frac{1}{(p^++k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times
$$

$$
\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\Big]
$$

$$
+ \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}\,(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}
$$

$$
\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right.
$$

$$
\left.- \frac{(x'-z)^{k'}}{(x'-z)^2}\,\Big[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right]\times
$$

$$
\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\Big]
$$

$$
- \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big] \times \Big[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\Big] \tag{4.26}$$

Row VI.

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Bigg[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left(\left[\frac{k^+}{p^+\,(p^+ + k^+)}\right]_+ + \frac{1}{(p^+ + k^+)}\right)\delta_{jm}\,\delta_{ik'}\Bigg]\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\Big]\\
&+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}\\
&\Big[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\Big]\Bigg[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\
&-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\Bigg]\times\\
&\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\Big]
\end{aligned} \tag{4.27}$$

In Rows III, V, VI the p^+ integrands fall like $\frac{1}{p^{+2}}$, so these rows are finite for $\vee\to\infty$.

Three ρ's. Group I Row II d^3N_{2b}, Rows III, IV, V (first part), Group II Row I (first part) and Group III Rows I, II, III (first part), V, VI have three ρ's. In the cross section they need $\langle\rho\rho\rho\rangle$, which is zero for a Gaussian projectile, or their two-ρ commutator parts ($[\rho^a(x),\rho^b(y)] = if^{abc}\rho^c(x)\delta^{(2)}(x-y)$).

4.9 The cross section

$$\begin{aligned}
\frac{d^3N_h}{d^3k} &= \int_{k^+/\vee}^1\frac{d\zeta}{\zeta^3}\,D_{h/g}(\zeta,\mu_F)\,\frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big)\qquad (d^3k = dk^+d^2\mathbf{k}),\\
\frac{d^3N_g}{d^3k} &= \mathrm{LO}\big|_{\alpha_s(\mu_R)} + \frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}\,\mathrm{LO} + \frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\mathrm{LO}\\
&\quad + \mathcal{N}\Big[\log\frac{k^+}{\Lambda}\,\mathrm{JIMWLK}\otimes\mathrm{LO} + \log\frac{\vee}{k^+}\,\mathrm{BFKL}\otimes\mathrm{LO}\Big]\\
&\quad + \frac{d^3N_{\rm rem}^{\rm G2\,IV}}{d^3k}\ (3.12) + \frac{d^3N_{V_\vee}}{d^3k}\ (4.2) + \text{the terms of Secs. 4.2–4.6, 4.8}\\
&\quad + \frac{\alpha_sN_c}{4\pi}\log\Big(1-\frac{k^+}{\vee}\Big)\mathrm{LO}\big[\hat E_X+\hat E_{X'}\big]\quad (\mathrm{Q1})\ +\ O(\alpha_s^3).
\end{aligned} \tag{4.28}$$

- The Jacobian is $\frac{1}{\zeta^3}$ for $d^3k = dk^+d^2\mathbf{k}$; by (1.2) this is $\frac{1}{\zeta^2}$ in the LO_ζ form. The $\frac{d\zeta}{\zeta^2}$ in `Remaining_RC_DGLAP_finite.pdf` (DGLAP and Result sections) and in the earlier finite-term notes (`Finite_terms_DGLAP_RC`, `_final`, `_singlet`, `_simplified`, `_exact`, `Comparison_AABKL`, `BFKL_origin_and_AABKL`) is $\frac{d\zeta}{\zeta^3}$; corrected in all of them. AABKL's $\frac{d\zeta}{\zeta^2}$ is right for $\frac{d\sigma}{d^2p\,d\eta}$.
- $\frac{d^3N_{\rm rem}^{\rm G2\,IV}}{d^3k}$ contains $-\log\frac{\vee}{k^+}H_4$, and the BFKL term of this row contains $+\log\frac{\vee}{k^+}H_4$.

5 (4) Running coupling

1. **Where β_0 appears in your file.** In two places only, both orange: the $\frac{11}{3}$ of Group I Row II part 1 (virtual) and the $-\frac{11}{6}$ of Group I Row V $d^3N_{2b\,\rm UV}$ (real, the two gluons of the loop at short distance $x\to z$). By (2.1) and (2.2),

$$\frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}\big[\hat E_X+\hat E_{X'}\big] - \frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}\big[\hat E_X+\hat E_{X'}\big] = 0 \tag{5.1}$$

exactly, for every $\vee$: the pole, γ_E, μ and the logarithm of the transverse size all cancel. With $y\to x$, $z\to w$ the two rows have the same measure, phase, kernel and colour bracket up to a sign, and $\frac{g^4N_c}{16\pi^4}\frac{11}{6} = \frac{g^4N_c}{32\pi^4}\frac{11}{3}$. So these rows produce **no running coupling**: no $\beta_0\log(r^2\mu^2)$ term with $r=|x-w|$ survives, so nothing in them fixes the scale of α_s at a transverse size of the emission.

2. **The coupling of LO.** LO is $\propto g^2$. In $\overline{MS}$, $g^2\to g^2(\mu_R)\big[1-\frac{\alpha_s\beta_0}{4\pi}(\frac{2}{\epsilon}+\log\frac{\mu^2}{\mu_R^2})\big]$ gives the counterterm (2.6). Its pole is the $\frac{\beta_0}{2}\delta(1-\xi)$ part of P_{gg} in the DGLAP pole (Sec. 3, Step 6). After $D_{h/g}$ is renormalized at μ_F, the only β_0 term left is the

renormalization-group term

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\beta_0} = \frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}\,\mathrm{LO}\big|_{\alpha_s(\mu_R)}.} \tag{5.2}$$

Since $\frac{d}{d\log\mu_R^2}\mathrm{LO}|_{\alpha_s(\mu_R)} = -\frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}$, the cross section does not depend on μ_R at this order. For $\mu_R = \mu_F$ the term is zero, and LO is taken with $\alpha_s(\mu_F)$.

3. **The constant.** The non-logarithmic part of Row II part 1 is (4.1), $\frac{\alpha_s}{2\pi}N_c(\frac{67}{9} - \frac{\pi^2}{3})\mathrm{LO}$. Here $N_c(\frac{67}{9} - \frac{\pi^2}{3}) = 2K$, where $K = N_c(\frac{67}{18} - \frac{\pi^2}{6})$ is the two-loop cusp constant ($n_f = 0$).
   - The constant depends on the scheme of α_s. Under $\alpha_s \to \alpha_s(1 + c\,\frac{\alpha_s}{2\pi})$ it shifts by $-c$.
   - It is absorbed completely by $\alpha_s \to \alpha_s(1 + 2K\frac{\alpha_s}{2\pi})$. That is twice the CMW shift $K\frac{\alpha_s}{2\pi}$ of soft-gluon emission, so here it is not the CMW scheme by itself. In $\overline{MS}$ it stays as a finite term.
4. **The $\frac{11}{6}$ in $d^3N_{2b\,\mathrm{finite}}$** has no pole and no μ. It is the UV counterterm subtracted from the rest of $2b$ (Sec. 4), not a running-coupling term. It must stay with the other lines of $d^3N_{2b\,\mathrm{finite}}$.
5. **Running coupling in the evolution.** The scale of α_s inside JIMWLK and BFKL (e.g. Balitsky's prescription) comes from the β_0 terms of the NLO kernels. They are of order $\alpha_s^2\delta Y$, beyond this calculation ($n_f = 0$, one emission). At this order, running coupling enters only through $\alpha_s(\mu_R)$ in LO and the renormalization-group term above.

6 Open point

Q1. The pole and μ that the counterterms do not remove:

$$\left.\frac{d^3N}{d^3k}\right|_{\mathrm{Q1}} = \frac{\alpha_s N_c}{4\pi}\log\Big(1 - \frac{k^+}{\vee}\Big)\mathrm{LO}\big[\hat{E}_X + \hat{E}_{X'}\big] \tag{6.1}$$

- It is neither DGLAP (not $P_{gg}\otimes\mathrm{LO}$) nor running coupling (no β_0).
- It comes from the cut $\xi \geq \frac{k^+}{\vee}$ in the virtual term of Group I Row II part 1, and vanishes for $\vee \to \infty$ at fixed k^+.
- With the virtual term integrated over $\int_0^1 d\xi$, as in AABKL (arXiv:1411.2869), both Q1 and $V_\vee$ are absent.

Reproducing

In `nlo_gluon_paper/other_check/`: `python3 diff_check.py` (your file against the checked list); `python3 checks.py` (μ table, prefactors, plus prescriptions, Fourier identity, convolution → `checks.json`); `python3 make_note.py` (this file).